Spec

Next step

- Add CITATION.CFF file to Git repos?

- DO WE HAVE A LANDING PAGE THAT POINTS TO ALL DOIs?

- Set up meeting with Bill, Alex, Emily, Chris, Juan, Stephen, and Dominic or Brendan
    - Where are we putting the DOI pages? How do they fit into the restructuring of the documentation?
    - Who is generating them?
    - Can they replace the existing specification pages?
    - Some meeting prep from Brendan (3/26)
        - Recommendations - trying to adhere to DataCite landing page documentation for DOIs + Generalist Repository Ecosystem Initiative recommendations
            - Position each assay schema as citeable DOI - full bibliographic citation at top of page
            - Human-readability as importance + clear areas for machine readability
                - Maybe at the top, list the following:
                    - Title
                    - Description
                    - DOI
                    - Version
                    - Publication date
                    - Publisher
                    - Contributors/contact
            - Clearly differentiate landing page vs. downloadable DOI object (the specific files)
            - GREI-specific thoughts
                - Include subjects for assays, even if minimal
                    - Dominic recommended either Ontology of Biomedical Investigations (OBI) or EDAM
                - Make reuse specific - e.g., “This schema may be reused, expanded, or referenced by external repositories”
    - Importance of abstracts
        - Description is the abstract
        - 4 types of description in DataCite
            - E.g., technicalAbstract

Info required for each DOI (from Bill)

- **Title**
  - dataset_type field values
- **Landing Page URL:**
  - pointer to a page as described below in the mockups
- **Description**
  - Can pull from here: https://docs.google.com/spreadsheets/d/1ZqJ78qEHr-W4JIAix8c5TFxeuPSJOJmO/edit?gid=251568642#gid=251568642
- **Creator:** "HuBMAP Consortium"
- **Resource type**: pick from the allowable value list under the "12.f resourceTypeGeneral" section here
  - Standard

DOI landing page content

DOI object

This is a zip file, downloadable from the landing page, containing the following files.

- **metadata.jsonld**
  - The CEDAR OpenView "Template JSON Schema" (see "Advanced View" at bottom of CEDAR page).
  - Pulled from here.
- **deprecated.json**
  - These are all of the deprecated fields, from the beginning of time.
  - Short description for the UI page: A cumulative list of field names that have been removed across all previous schema versions and are no longer part of the current schema.
- **metadata.yml**
  - A YAML version of the metadata schema.
  - Pulled from here.
- **metadata.xlsx**
  - The "Excel template" HuBMAP distributes in the Data Upload Guidelines pages (histology example)
- **metadata.tsv**
  - The "TSV template" HuBMAP distributes in the Data Upload Guidelines pages (histology example)
- **directory.md**

    - This is a markdown file containing the following three sections:
        - The “Directory schemas” table from the Data Upload Guidelines (histology example)
        - The directory schema represented as a tree (this is the regexp, not an example)
        - The YAML file used to define the directory structure (histology example)
    - *For “legacy” we need to use the legacy directory structure for DOIs pointing to legacy schemas*
- **empty_tree.zip**
    - The directory structure as a zip’d folder structure
    - Not sure if this would be truly platform independent but the idea is if someone unzips the file then they have the schema-appropriate (empty) directory tree

Here is an example of the “DOI object” (i.e., zip file) for histology.

Optional content we might want to include in the DOI object:

- DOI descriptor file
    - If our DOIs are autogenerated, there will likely be a descriptor file (yaml?) that’s created per assay type. If useful, we could include this descriptor file in the ZIP.

Required content on the page

This is in addition to the downloadable DOI zip file.

- Title (i.e., dataset type?)
- DOI
- Description
- Contributor(s) / ~~Contact~~
    - Can use “Creator” instead of “Contributors” → “HuBMAP Consortium”
- Date published
- Version
    - ~~“Legacy” is an option for the case where a schema existed before our major restructuring. The legacy will include both the metadata and file schemas.~~
    - Start with “Version 1” as static text. Add a link to new versions, when relevant.
    - Add link to “Unified View of Metadata” (Bill’s harmonized table)
- Link to CEDAR Validator

- Link to CEDAR OpenView metadata page and/or include content in DOI page (Mockup 1)
    - CAN'T LINK TO OPENVIEW BECAUSE CATEGORICAL VALUES CAN CHANGE
    - Need to regenerate openview documentation with links to BioPortal removed
        - Might be able to get this from Bill's group (via unified view efforts)
    - There is an additional section where deprecated fields are detailed.
    - Add tool tip for "deprecated"
- File structure regexp table and example tree (Mockup 2)
- Link to reference describing standards (DCWG paper)
- Descriptions of the files in the DOI object (i.e., ZIP). The descriptions are here.
- MD5 hash of DOI zip file
- Link to HuBMAP/SenNet data portal filtered by datasets using this specification.
    - Some assays will be consortium-specific. In those cases we just point to the appropriate data portal.
    - For assays that are shared, we should include both SenNet and HuBMAP buttons on the same DOI page, pointing to their respective data portals.
    - Here is an prototype for HuBMAP
        - https://portal-prod.test.hubmapconsortium.org/search/datasets?dataset_type=CODEX
        - Note that multi-word dataset types such as "10X Multiome" need to be provided as "10X+Multiome" in the url parameter you described.
        - Dataset type capitalization needs to match the actual values. i.e. "HISTOLOGY" will not work, but "Histology" will. Nils' team can look into making it case insensitive if that is needed.
- DOI should be (1) included in html metadata and (2) displayed as a URL, per DataCite.
- (optional) "Subject" - a text field that contains standards-specific ontological keywords. For histology this could be the stains.
    - Not sure where or how we'll maintain this list for each assay.

Example DOI landing page

Mockup 1

The **first tab** would show the metadata schema documentation, pulled from CEDAR OpenView.

HuBMAP

Metadata Reporting Standards

Histology

The microscopic study of tissue composition and structure, often referred to as microscopic anatomy. It involves examining tissue samples, typically after they've been sectioned, stained, and placed under a microscope.

Version 1

?

Download (ZIP)

Published: September 26, 2025

https://doi.org/10.35079/HBM788.QPBW.699

MD5 hash:
5d41402abc4b2a76b9719d911017c592

Subjects: AB-PAS, H&E, H-DAB, LFB, PAS, SBB, Trichrome

Descriptive Metadata | File Organization

Schema

Parent sample ID *

Unique HuBMAP or SenNet identifier of the sample (i.e., block, section or suspension) used to perform this assay. For example, for a RNAseq assay, the parent would be the suspension, whereas, for one of the imaging assays, the parent would be the tissue section. If an assay comes from multiple parent samples then this should be a comma separated list. Example: HBM386.ZGKG.235, HBM672.MKPK.442 or SNT232.UBHJ.322, SNT329.ALSK.102

Lab ID

An internal field labs can use it to add whatever ID(s) they want or need for dataset validation and tracking. This could be a single ID (e.g., "Visium_9OLC_A4_S1") or a delimited list of IDs (e.g., "9OL; 9OLC.A2; Visium_9OLC_A4_S1"). This field will not be accessible to anyone outside of the consortium and no effort will be made to check if IDs provided by one data provider are also used by another.

Preparation protocol DOI *

DOI for the protocols.io page that describes the assay or sample procurement and preparation. For example for an imaging assay, the protocol might begin with staining of a section and finalize with the creation of an OME-TIFF file. In this case the protocol would include any image processing steps required to create the OME-TIFF file. Example: https://dx.doi.org/10.17504/protocols.io.eq2lyno9qvx9/v1.

Dataset type *

The specific type of dataset being produced.

Histology - (https://purl.humanatlas.io/vocab/hravs#HRAVS_0000197)

Analyte class *

Analytes are the target molecules being measured with the assay.

Deprecated

Tiled image columns #

The number of columns used in the stitching process of a tiled image, often referred to as the grid size in the x-dimension. Example: 5

Tiled image count #

The total number of raw tiled images captured, which are intended to be stitched together. Example: 75

CEDAR OpenView Documentation

CEDAR Validator

HuBMAP Datasets

The HuBMAP Data Portal is an open platform to discover, visualize and download standardized healthy single-cell and spatial tissue data.

Mockup 2

The **second tab** would show the file hierarchy schema table that's currently in our GitHub documents and the directory schema displayed as a simple tree. Both of these are also included in the directory.md file that's in the DOI zip.

Metadata Reporting Standards

Histology

The microscopic study of tissue composition and structure, often referred to as microscopic anatomy. It involves examining tissue samples, typically after they've been sectioned, stained, and placed under a microscope.

Version 1

?

Download (ZIP)

Published: September 26, 2025

https://doi.org/10.35079/HBM788.QPBW.699

MD5 hash:
5d41402abc4b2a76b9719d911017c592

Subjects: AB-PAS, H&E, H-DAB, LFB, PAS, SBB, Trichrome

Descriptive Metadata | File Organization

Details

pattern	required?	description
extras\/.*	✓	Folder for general lab-specific files related to the dataset.
extras\/microscope_hardware\.json	✓	**[QA/QC]** A file generated by the micro-meta app that contains a description of the hardware components of the microscope. Email HuBMAP Consortium Help Desk help@hubmapconsortium.org if help is required in generating this document.
extras\/microscope_settings\.json		**[QA/QC]** A file generated by the micro-meta app that contains a description of the settings that were used to acquire the image data. Email HuBMAP Consortium Help Desk help@hubmapconsortium.org if help is required in generating this document.
raw\/.*	✓	Raw data files for the experiment.
raw\/images\/.*	✓	Raw image files. Using this subdirectory allows for harmonization with other more complex assays, like Visium that includes both raw imaging and sequencing data. This directory should include at least one raw file.

Example

```
.
├── metadata.tsv
├── extras/
│   ├── contributors.tsv
│   └── microscope_hardware.json
└── raw/
    ├── images/
    │   ├── Brightfield_HE_10msec_image_0001.tiff
    │   ├── Brightfield_HE_10msec_image_0002.tiff
    │   ├── ...
    │   └── Brightfield_HE_10msec_image_0110.tiff
    └── lab_processed/
        └── images/
            ├── Histology_13_95.ome.tiff
            ├── Histology_13_95.channels.csv
            └── S1.tissue-boundary.geojson
```

HuBMAP Datasets

The HuBMAP Data Portal is an open platform to discover, visualize and download standardized healthy single-cell and spatial tissue data.

Mockup 3

This is the content for the help screen (i.e., the “?” button next to “Download”. It contains descriptive text about each of the files inside the zip archive. This is an incomplete mockup, for the full descriptions, see here.

Metadata Reporting Standards

Documents Provided

The HuBMAP Metadata Reporting Standards include schemas for both descriptive and structural metadata. The descriptive metadata provides a standardized description of samples, experiments, and processing context. The structural metadata provide standardized file organization including hierarchical file and directory specifications, file type and naming conventions

The metadata standards vary for different types of experiments, although there is significant overlap across standards. These standards are reported using five files: JSON (metadata), XLSX (metadata), TSV (metadata), MD (files and directories), ZIP (directories)

metadata.json (JSON file)

JSON Schema Description

This JSON schema defines the structure and validation rules for metadata instances generated from a metadata template in the CEDAR framework. The schema is automatically produced by the CEDAR Workbench and can be accessed through the template's OpenView page.

The schema follows the JSON Schema specification and includes a JSON-LD context to provide semantic meaning for metadata fields. It defines the set of properties, required fields, data types, and value constraints used to validate metadata instances.

Each schema contains:

- Template metadata (name, identifier, version)
- Field and element definitions describing the metadata structure
- Validation constraints such as required fields and controlled values
- UI configuration used to render metadata entry forms

The metadata schema supports the standardized description of HuBMAP datasets by defining a consistent structure for dataset metadata. It enables validation of submitted metadata, promotes interoperability across HuBMAP systems, and facilitates reliable data discovery and reuse.

metadata.xlsx (Microsoft Excel file)

Microsoft Excel File Description

This is placeholder text.
See the Google Doc for the full content.

DOI FILE DESCRIPTIONS

This page contains descriptions of the files in the deliverables ZIP archive. These descriptions are expected to be included in a help window, accessed from the DOI landing pages.

metadata.jsonld (JSON-LD file)

JSON Metadata Schema Description

This JSON metadata schema defines the structure and validation rules for descriptive metadata instances generated from a metadata template in the CEDAR framework. The schema is automatically produced by the CEDAR Workbench and can be accessed through the template's OpenView page.

The schema follows the JSON Schema specification and includes a JSON-LD context to provide semantic meaning for metadata fields. It defines the set of properties, required fields, data types, and value constraints used to validate metadata instances.

Each schema contains:

- Template metadata (name, identifier, version)
- Field and element definitions describing the metadata structure
- Validation constraints such as required fields and controlled values
- UI configuration used to render metadata entry forms

The metadata schema supports the standardized description of HuBMAP datasets by defining a consistent structure for dataset metadata. It serves as the primary machine-readable representation for validation, exchange, and automated processing.

deprecated.json (JSON file)

JSON List

The deprecated file tracks all field names that have been retired throughout the schema's history. When a field is removed in any schema version, it is added to this file, making it an accumulative record across all past versions, not just the most recent one. This allows consumers of the schema to identify legacy fields that may still exist in older data but are no longer recognized or supported by the current schema definition.

metadata.yml (YAML file)

YAML Metadata Schema Description

A simplified YAML version of the descriptive metadata schema, provided to make the schema easier to read and interpret by humans, while still preserving a structured format that can support computational use.

metadata.xlsx (Microsoft Excel file)

Excel Spreadsheet Template Description

The Excel spreadsheet template is generated from a CEDAR metadata template to support bulk metadata submission by data providers. It provides a table-based format for collecting metadata across multiple records. In addition to the main data-entry sheet, the workbook includes supporting sheets that list permitted values and standardized term identifiers. These supporting sheets serve as the source of the dropdown options available in the data-entry sheet, helping users select valid terms consistently, reduce manual entry errors, improve data quality, and make bulk submission easier and faster.

The Excel spreadsheet template is compatible with the [Metadata Spreadsheet Validator](https://metadatavalidator.metadatacenter.org) (https://metadatavalidator.metadatacenter.org), which can be used to check the correctness of bulk metadata records before submission.

metadata.tsv (tab-separated values file)

Tab-Separated Values (TSV) Template Description

The TSV file template is generated from a CEDAR metadata template to support bulk metadata submission in a simple tab-delimited format. It provides a table-based structure for collecting metadata across multiple records, with one row per record, and with each column corresponding to a metadata field in the template. The TSV version is intended for users and systems that prefer a lightweight, machine-readable format for data entry, exchange, and automated processing.

The TSV template is compatible with the [Metadata Spreadsheet Validator](https://metadatavalidator.metadatacenter.org) (https://metadatavalidator.metadatacenter.org), which can be used to check the correctness of bulk metadata records before submission.

directory.md (markdown file)

Markdown Structural Metadata Description

The markdown file details the structural metadata schema specification. The file includes three sections:

- File Organization Schema
    - This section includes a table that contains the file organization schema (i.e., hierarchical file and directory specifications, including file type and naming conventions). This table documents each element in the schema, providing descriptions and denoting if the file or directory is required. Regular expressions are used to detail each of the schema elements.
- Directory Tree
    - The file hierarchy schema is represented as a directory tree. The directory tree offers a more human-readable view of the schema.
- YAML
    - This section details the file organization schema in a YAML structure. This YAML code can be used with the HuBMAP Ingest Validation Tools to validate a data-filled directory tree.

empty_tree.zip (ZIP archive)

ZIP Archive Description

The ZIP archive contains an empty directory structure (i.e., devoid of any files). The ZIP archive preserves the metadata standards' hierarchical folder layout. When unzipped, it will create the appropriate dataset-specific directory tree on the destination system. This directory tree can be used as a framework for structuring data files to meet the dataset-specific file hierarchy standard.

Notes

Notes

May 6, 2026 | Dataset DOI's

Attendees: Philip Blood Chris Csonka Emily Nic Juan Muerto Fisher, Stephen A Shirey, Bill Alex Ropelewski Brendan Honick

[Mockup]

Action items

- [ ] Where are we putting the DOI pages?
    - [ ] How do they fit into the restructuring of the documentation?
        - [ ] **Don't need to fit, will linked between two**
    - [ ] **Put in directory under documentation site on GitHub**
- [ ] Who is generating the DOI objects (Zip)?
    - [x] ~~Juan's team~~
        - [ ] The zip objects are located [here]
- [ ] Who is creating the (static) DOI web pages?
    - [ ] Jess will generate schema description content
    - [ ] Bill's team generate template
    - [ ] Stephen helping collate all datatype details - generate master spreadsheet
    - [ ] Chris create hard-coded tables from template + collated details
- [ ] Is there a landing page that will point to all of the DOIs (would be helpful to cite in paper)?
    - [ ] "Collection DOI" - create a page for the collection
- [ ] What is the timeline for creation?
    - [ ] Bill's team - end of May to June 8th
    - [x] ~~Juan's team - before May 29th~~

Notes

Mar 26, 2026 | HuBMAP DOIs for Assays/datatypes

Attendees: Chris Csonka Brendan Honick Alex Ropelewski Juan Muerto Dominic Bordelon Fisher, Stephen A Emily Nic ~~Shirey, Bill~~

Notes

- DOI needs to point to an object, which the team decided would be a zip file

- DCWG paper will be used for citation for DOI object

Action items

- [ ] Next steps: discuss with Bill (wasn’t able to attend this meeting) - Stephen, Brendan, Chris, Juan, CC Dominic and Alex